Jose Luis Muñoz Zuta

Payments Software Engineer | Card Transaction Processing: Switching, Clearing & Settlement

Lima, Peru (open to remote) | +51 987 570 658 | joseluismunozzuta@gmail.com | linkedin.com/in/jlmz

SUMMARY

Payments software engineer with 6+ years in transactional banking, specialized in the end-to-end card transaction lifecycle: authorization switching, clearing with Visa and Mastercard, and merchant settlement and reconciliation. Deputy technical lead at FIS for the production processing platform of **Culqi (BCP)**, one of Peru's leading payment processors (**~2M transactions per cycle, same-day merchant settlement**). Proven track record in batch performance optimization, card scheme compliance and interchange qualification, and leading the resolution of critical production incidents. Hands-on with C/C++, Pro*C, Java/Spring Boot, Python and Oracle SQL; as a freelancer, also built and maintain a Spring Boot property management system in production across a 5-hostel chain.

SKILLS

Payments: Card transaction lifecycle (authorization, clearing, settlement), Visa & Mastercard interchange qualification, card scheme mandates & compliance, merchant settlement & reconciliation, ISO 8583, EMV, FIS IST (Switch, Clearing, MAS)

Programming: C, C++, Java, Python, SQL, Pro*C, Shell scripting, JavaScript / Node.js

Backend & Data: Spring Boot, REST APIs, microservices, Oracle Database, SQL performance tuning

Core strengths: Production incident leadership, batch performance optimization, product release customization & deployment, process automation, client and card-brand communication

PROFESSIONAL EXPERIENCE

FIS · Lima, Peru · Remote — Apr 2023 – Present

Senior Consultant — Jul 2025 – Present

Deputy technical lead for the production card processing platform of Culqi (BCP), built on FIS IST Switch, Clearing and MAS: ~2M transactions per cycle, 3 cycles per day, same-day merchant settlement. Involved since project inception, through Culqi's growth into one of Peru's reference payment processors.

- **Reduced Clearing batch processing time by 60% (25 → 10 min)** to absorb Culqi's transaction growth within the 2-hour window between cycle cut-off and merchant payout, through in-memory data loading, SQL query and index optimization, and server memory capacity planning.
- **Ensured accurate Visa and Mastercard interchange fee assignment for the Peruvian market** by designing custom fee-qualification rules for domestic programs (card-present, e-commerce and EMV indicators), extending a product built for international programs only.
- **Lead the resolution of critical production incidents** across the Switch → Clearing → MAS flow, defining the recovery strategy (file regeneration, reprocessing or resubmission) while preventing duplicate submissions to the card brands.
- **Serve as technical liaison with Visa and Mastercard**, together with the client, to resolve clearing file rejections and implement scheme programs such as Micromerchant reduced interchange rates.
- **Own the customization and deployment of FIS product releases carrying card scheme mandates**, aligning each upgrade with business requirements while preserving existing customizations and production stability.
- **Develop settlement and reconciliation reporting** (payment orders, approved and suspended transactions, control totals) that enables same-day payment to merchants.
- **Automated high-volume data operations with Python**, generating scheduled shell jobs and bulk SQL INSERT/UPDATE scripts for data corrections, reprocessing and merchant configuration.

Tech: C, C++, Pro*C, Java, Spring Boot, Python, Oracle SQL

Analyst Programmer II — Apr 2023 – Jul 2025

- Contributed to the Culqi (BCP) implementation from inception, developing custom clearing and merchant settlement logic (C, C++, Pro*C) for the Peruvian market.
- Developed merchant onboarding REST APIs (Java/Spring Boot) to register and update merchants and tax IDs (RUC) in the processing platform, with built-in business validations.
- Delivered Switch and Clearing customizations for Latin American processors Redeban / Bre-B (Colombia) and E-Global (Mexico).

Hostales Elixer · Lima, Peru · Freelance — Apr 2022 – Present

Freelance Software Engineer (sole developer)

- **Designed and built from scratch the property management system that runs daily operations across Hostales Elixer's 5 hostels in Lima**: room status, check-in/check-out, extra hours, guest registry, product sales, inventory and shift closing. In production since 2022 under an annual support and maintenance license.
- Java/Spring Boot backend (Spring Security, Spring Data JPA, MySQL) with Thymeleaf/Tailwind UI and real-time notifications via WebSockets and a Telegram bot.
- Automated management reporting: revenue and occupancy dashboards, PDF/Excel exports and end-of-shift reports emailed to the owners; packaged as a one-click Windows installer (embedded Java runtime, MySQL and the app running as Windows services) for on-premise deployment.

  Tech: Java, Spring Boot, Spring Security, JPA/Hibernate, MySQL, Thymeleaf, WebSockets, Apache POI, OpenPDF

Oktana · Lima, Peru — Dec 2021 – Feb 2023

Software Engineer (Salesforce)

- Developed and deployed Salesforce applications for client projects using Apex, Lightning Web Components and process automation.
- Front-end developer on a Dreamforce demo project for Salesforce.
- Earned three Salesforce certifications within one year: Platform Developer I & II and JavaScript Developer I.

Novatronic · Lima, Peru — Jan 2020 – Dec 2021

Software Engineer (Aug – Dec 2021) · C Developer Intern (Jan 2020 – Aug 2021)

- Developed C modules for banking transaction switches processing e-commerce, ATM and POS payments (ISO 8583, EMV, information security).
- Performed unit and integration testing and software certification to validate releases against client requirements.

EDUCATION

Pontificia Universidad Católica del Perú (PUCP) · Lima, Peru — 2016 – 2021

B.Sc. in Electronic Engineering · Professional title of Electronic Engineer (2023)

- Top 10% of class (Décimo superior). Thesis research with the IoT-PUCP Research Group.

Université de Montréal · Montréal, Canada — Aug – Dec 2019

Exchange semester in Computer Science

- Emerging Leaders in the Americas Program (ELAP) scholarship, Government of Canada; DARI PUCP scholarship.

RESEARCH PROJECT

Intelligent in-home and remote monitoring system for older adults living alone (undergraduate thesis)

- Designed and implemented an end-to-end Ambient Assisted Living system that monitors an older adult's heart rate in real time and **sends instant emergency alerts to family members**; validated in a real home.
- Wearable with **93% heart-rate accuracy** and fall detection, wireless emergency buttons, fall-prevention alerts, an ESP32 control unit with GSM/SMS backup, and a Firebase-backed Android app.

CERTIFICATIONS

- Salesforce Platform Developer II (2023) · Platform Developer I (2022) · JavaScript Developer I (2022)
- Node.js Backend Development – Coderhouse (2023)
- DELF B2 – France Éducation international

LANGUAGES

Spanish: native · **English:** advanced (full professional) · **French:** DELF B2 · **Portuguese:** advanced